

MACHINE LEARNING FUNDAMENTALS

Understanding How Machine Learning Models Learn

Inquiry-Based Learning Report

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Assignment	Questions About Training Methods
Artifact	Professional Portfolio Artifact 3

This report reorganizes the guided conversation into a professional learning artifact while preserving the concepts, examples, and conclusions developed during the inquiry.

1. Introduction

This artifact documents an inquiry-based learning activity about how machine learning models are trained. The discussion explored supervised, unsupervised, and reinforcement learning, as well as algorithms, parameter adjustment, repetition, data quality, and overfitting. The activity helped me connect technical concepts with practical examples and explain how models improve through training and evaluation.

2. Supervised Learning

A supervised learning model learns from labeled training data. Each example includes an input and the correct output. The model makes a prediction, compares that prediction with the correct answer, measures the error, and adjusts its internal parameters. This process is repeated until the model becomes more accurate.

The term supervised refers to the fact that correct answers guide the learning process. For example, a model trained to recognize cats would receive images together with labels indicating whether each image contains a cat.

3. How the Model Adjusts Its Parameters

The model uses a feedback process to determine how much its internal parameters should change after an incorrect prediction.

- **Loss Function:** Measures how wrong the prediction is. A larger error produces a higher loss, while a smaller error produces a lower loss.
- **Gradient:** Shows the direction in which each parameter should move to reduce the loss.
- **Learning Rate:** Controls the size of each adjustment so the model improves gradually instead of making random or overly large changes.

Together, the loss function, gradient, and learning rate allow the model to improve step by step.

4. Unsupervised Learning

In unsupervised learning, the model receives unlabeled data and identifies patterns, similarities, or groups without being given correct answers.

A retail company could use customer purchase data and allow the model to group customers according to similar buying habits. The model might identify frequent shoppers, high spenders, or occasional customers even though those categories were not defined in advance.

Human judgment is still needed to interpret what the discovered groups mean in a real-world context.

5. Reinforcement Learning

In reinforcement learning, an agent learns by interacting with an environment and receiving rewards or penalties for its actions. It tries different actions, observes the results, and gradually learns which choices produce the highest total reward.

For example, a warehouse robot may receive a reward for reaching the correct location, a penalty for hitting an obstacle, and a smaller penalty for taking too long. Through repeated attempts, it learns a safer and more efficient route.

The agent must balance exploration and exploitation. Exploration means trying new actions, while exploitation means using actions that have worked well before.

6. Why Algorithms Matter

Algorithms provide the rules and mathematical steps a machine learning model uses to learn from data. They determine how the model identifies patterns, measures errors, updates its parameters, and makes predictions.

Different algorithms are appropriate for different types of problems. Linear regression can predict a numerical value such as a house price, while a decision tree can classify an email as spam or not spam.

The algorithm can be understood as the model's learning strategy. The data provides the examples, but the algorithm determines how the model learns from those examples.

7. Basic Steps in Training a Machine Learning Model

1. Define the problem.
2. Collect relevant data.
3. Clean and prepare the data.
4. Split the data into training, validation, and testing sets.
5. Choose an appropriate algorithm.
6. Train the model.
7. Evaluate model performance.
8. Improve the model by adjusting settings or improving the data.
9. Deploy the model for real-world use.
10. Monitor the model after deployment because new data may differ from the training data.

Evaluation and monitoring are especially important because a model may perform well during training but still struggle with new or changing data.

8. Repetition and Epochs

Repetition helps a model improve gradually. During training, the model may pass through the same dataset several times. Each full pass through the data is called an epoch.

During every epoch, the model makes predictions, measures errors, and adjusts its parameters. One pass may not be enough to learn the important patterns, so repeated passes provide additional opportunities to reduce mistakes.

Too much repetition can cause overfitting. Validation data helps determine when the model should stop training.

9. The Role of Training Data

Examples, or data, are the foundation of machine learning because models learn patterns from the information they receive.

- **Supervised learning:** uses inputs and correct labels.
- **Unsupervised learning:** uses unlabeled examples to discover patterns.
- **Reinforcement learning:** uses actions, experiences, rewards, and penalties.

The quality of the data directly affects the quality of the model. If the examples are inaccurate, biased, incomplete, or not representative of real situations, the model may make unreliable predictions.

For example, a facial recognition model trained only on clear, brightly lit images may struggle with darker or lower-quality images. Training data should therefore be accurate, diverse, relevant, and representative. More data can help, but high-quality data is more important than quantity alone.

10. Understanding Overfitting

Overfitting occurs when a model memorizes the specific details of the training data, including noise, instead of learning broader patterns that work on new data.

Recognizing Overfitting

The clearest sign is a large gap between training performance and validation or testing performance.

- High training accuracy.
- Low validation or testing accuracy.
- Training error continues to decrease while validation error begins to increase.

Preventing Overfitting

- **More representative data:** Additional varied examples make memorization more difficult and encourage generalization.
- **Simpler models:** Reducing unnecessary complexity can prevent the model from learning overly specific rules.
- **Regularization:** Adds a penalty for complexity and discourages the model from depending too heavily on individual features.
- **Early stopping:** Stops training when validation performance begins to decline even though training performance continues to improve.
- **Validation monitoring:** Tracks performance on unseen data during training.

High training accuracy alone does not prove that a model is effective. A useful model must generalize well to new data.

11. Comparison of Learning Approaches

Approach	Training Signal	Main Goal	Example
Supervised Learning	Labeled examples and correct answers	Learn to predict known outputs	Cat image classification
Unsupervised Learning	Unlabeled data	Discover patterns, similarities, or groups	Customer segmentation
Reinforcement Learning	Rewards and penalties from interaction	Learn actions that maximize total reward	Warehouse robot navigation

12. Lessons Learned

This activity gave me a clearer understanding of how machine learning models learn and improve. I learned that model training is a gradual process based on data, prediction, error measurement, and parameter adjustment.

I also learned that supervised, unsupervised, and reinforcement learning use different types of feedback. Algorithms act as learning strategies, repetition creates opportunities for improvement, and data quality strongly affects the reliability of the final model.

The discussion of overfitting was especially important because it showed that excellent training performance can be misleading. Validation, testing, monitoring, and generalization are necessary to determine whether a model will work in real situations.

13. Conclusion

This inquiry activity strengthened my understanding of how machine learning models learn from data. Supervised learning uses labeled examples, unsupervised learning discovers patterns in unlabeled data, and reinforcement learning improves through rewards and penalties.

Algorithms, loss functions, gradients, learning rates, repetition, and data quality all influence model performance. Most importantly, high training accuracy does not guarantee success. A useful model must generalize well to new data and should be monitored for problems such as overfitting.

Appendix: Original Inquiry Topics

- How does a supervised learning model learn from training data?
- How does a model know how much to adjust its parameters?
- What is the main approach used in unsupervised learning?
- How does reinforcement learning work?
- Why are algorithms important?
- What are the basic steps in training a model?
- How does repetition help?
- What role does data play?
- How can overfitting be recognized and prevented?